

How to Run CORBA Calculator Program

Java Distributed Systems — Assignment 2 — Practical Exam Guide

CORBA vs RMI — Key Difference (remember for exam!)

	RMI (Assignment 1)	CORBA (Assignment 2)
Language	Java only	Any language (Java, C++, Python...)
Contract file	Java interface (.java)	IDL file (.idl)
Middleman	RMI Registry	ORB + Naming Service
Extra step	None	Must compile IDL first (idlj)
Port used	1099	1050 (you choose)

STEP 0 — Keep All Files in ONE Folder

You need these 3 files in the same folder:

```
Calculator.idl CalculatorServer.java CalculatorClient.java
```

STEP 1 — Open Command Prompt and Go to Your Folder

Press Win+R → type cmd → Enter. Then:

```
cd "C:\Users\Srushti Nitin Pagar\Desktop\PPA Home\DS"
```

STEP 2 — Compile the IDL File (generates Java files automatically)

CORBA needs this EXTRA step that RMI does not! The IDL compiler reads Calculator.idl and auto-generates Java helper files.

```
idlj -fall Calculator.idl
```

■ This creates a new folder called 'CalculatorApp' with auto-generated .java files inside.

■ Do NOT edit those auto-generated files — they are created by the tool.

STEP 3 — Compile All Java Files

Compile everything at once (includes auto-generated files):

```
javac *.java CalculatorApp/*.java
```

■ No output = SUCCESS. Many .class files will be created.

STEP 4 — Open THREE Terminal Windows

CORBA needs 3 terminals (unlike RMI which needs 2):

Terminal	Purpose
Terminal 1	Run the CORBA Naming Service (like RMI Registry)
Terminal 2	Run the Calculator Server
Terminal 3	Run the Calculator Client

Open 3 CMD windows: Press Win+R → type cmd → Enter (do this 3 times)

In ALL 3 terminals, navigate to your folder:

```
cd "C:\Users\Srushti Nitin Pagar\Desktop\PPA Home\DS"
```

STEP 5 — Start the Naming Service (Terminal 1) — Run FIRST!

The Naming Service is CORBA's phone book. Start it on port 1050:

Option A (Standard):

```
orbd -ORBInitialPort 1050
```

Option B (If Option A doesn't work):

```
tnameserv -ORBInitialPort 1050
```

■ You'll see some startup messages — that's normal. Keep this window open!

STEP 6 — Start the Server (Terminal 2) — Run SECOND!

Option A (Standard):

```
java CalculatorServer -ORBInitialHost localhost -ORBInitialPort 1050
```

Option B — With classpath (if ClassNotFoundException):

```
java -cp . CalculatorServer -ORBInitialHost localhost -ORBInitialPort 1050
```

■ You should see: Calculator Server is running...

■ Keep this window open! Do NOT close it.

STEP 7 — Run the Client (Terminal 3) — Run LAST!

Option A (Standard):

```
java CalculatorClient -ORBInitialHost localhost -ORBInitialPort 1050
```

Option B — With classpath (if ClassNotFoundException):

```
java -cp . CalculatorClient -ORBInitialHost localhost -ORBInitialPort 1050
```

■ Expected output:

```
Addition (10 + 5): 15.0 Subtraction (10 - 5): 5.0 Multiplication (10 * 5):  
50.0 Division (10 / 2): 5.0 Error: Division by zero is not allowed!
```

Quick Reference — All Commands in Order

#	Command	Terminal
1	idlj -fall Calculator.idl	Any (once only)
2	javac *.java CalculatorApp/*.java	Any (once only)
3	orbd -ORBInitialPort 1050	Terminal 1 — keep open
4	java -cp . CalculatorServer -ORBInitialHost localhost -ORBInitialPort 1050	Terminal 2 — keep open
5	java -cp . CalculatorClient -ORBInitialHost localhost -ORBInitialPort 1050	Terminal 3

Common Errors & Fixes

Error	Fix
ClassNotFoundException	Use -cp . before class name
Cannot find NameService	Start orbd/tnameserv FIRST (Step 5)
Address already in use: 1050	Naming service already running — skip to Step 6
idlj not found	Try: %JAVA_HOME%\bin\idlj -fall Calculator.idl
CalculatorApp folder missing	Run idlj step again (Step 2)
Division by zero error shown	CORRECT behavior — exception from server works!

Exam Remember: IDL compile → Java compile → Naming Service → Server → Client
CORBA needs 3 terminals! RMI needs only 2.